

CSTM Analysis R IVS

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Abstract

This document summarises an analysis of the European Values Study. In this project, I analyse the relationship between environmental and phenotypical factors, and personality traits.

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```
### Load the Stata dataset ###
data <- read_dta(here("ZA7503_v3-0-0.dta"))
t <- data
```

1 First steps

1.1 The European Values Study

The European Values Study database includes 224434 observations and 635 variables.

1.2 Define the research question

What is the relationship between resource availability and personality traits? To what extent does health mediate this relationship?

1.3 Define the prediction(s)

- 1) Resource availability is linked to a higher level in each personality trait.
- 2) Health partly mediate the relationship between resource availability and personality traits.

```
### Keep relevant variables in ts ###
ts <- t[, c("S007", "S009", "S017", "S020",
            "Y001",
            "A010", "A058", "A059", "A061", "A063", "A118", "C025", "E049",
            "A066", "A071", "A083", "A088", "E033", "E046", "E061", "E104", "E108",
            ↪ "E143", "F001",
            "A064", "A080_01", "A081", "A107", "A108", "A168A", "B001",
            "A119", "C041", "C042B5", "E048", "E051",
            "A017", "A019", "A173", "E047", "F010",
            "A009", "X001", "X003", "X007", "X025", "X025R", "X028", "X045", "X047_EVS",
            "F024")]

# S007 (id), S009 (country_abb), S017 (weight), S020 (survey_year)
# Y001 (postmat)
# A010 (ext_excited), A058 (ext_no_friends), A059 (ext_no_colleagues), A061
↪ (ext_no_community), A063 (ext_no_persuasion), A118 (ext_v_social), C025
↪ (ext_meeting), E049 (ext_convince)
# A066 (ope_arts), A071 (ope_ecology), A083 (ope_v_arts), A088 (ope_v_ecology), E033
↪ (ope_no_liberal), E046 (ope_new), E061 (ope_reform), E104 (ope_no_app_ecology), E108
↪ (ope_no_app_women), E143 (ope_no_immigration), F001 (ope_never_meaning)
# A064 (agr_welfare), A080_01 (agr_humanitarian), A081 (agr_v_welfare), A107
↪ (agr_v_solidarity), A108 (agr_v_compassion), A168A (trust), B001 (agr_no_give)
# A119 (con_v_skills), C041 (con_no_work), C042B5 (con_imp_work), E048 (con_success),
↪ E051 (con_get)
# A017 (sta_depressed), A019 (sta_upset), A173 (sta_control), E047 (sta_serenity), F010
↪ (sta_no_meaning)
# A009 (health), X001 (sex), X003 (age), X007 (marital_status), X025 (education), X025R
↪ (education_levels), X028 (employment_status), X045 (social_class), X047_EVS
↪ (income_scale)
# F024 (religion)
```

```

### Give new names to variables in ts ###
names(ts) <- c(
  "id", "country_abb", "weight", "survey_year",
  "postmat",
  "ext_excited", "ext_no_friends", "ext_no_colleagues", "ext_no_community",
  "ext_no_persuasion", "ext_v_social", "ext_meeting", "ext_convince",
  "ope_arts", "ope_ecology", "ope_v_arts", "ope_v_ecology", "ope_no_liberal", "ope_new",
  "ope_reform", "ope_no_app_ecology", "ope_no_app_women", "ope_no_immigration",
  ↪ "ope_never_meaning",
  "agr_welfare", "agr_humanitarian", "agr_v_welfare", "agr_v_solidarity",
  ↪ "agr_v_compassion", "trust",
  "agr_no_give",
  "con_v_skills", "con_no_work", "con_imp_work", "con_success", "con_get",
  "sta_depressed", "sta_upset", "sta_control", "sta_serenity", "sta_no_meaning",
  "health", "sex", "age", "marital_status", "education", "education_levels",
  ↪ "employment_status",
  "social_class", "income_scale",
  "religion"
)

### Reorder the columns in ts ###

# Define the order of column names
column_order <- c(
  "id", "sex", "age", "weight", "country_abb", "survey_year",
  "ext_excited", "ext_no_friends", "ext_no_colleagues", "ext_no_community",
  "ext_no_persuasion", "ext_v_social", "ext_meeting", "ext_convince",
  "ope_arts", "ope_ecology", "ope_v_arts", "ope_v_ecology", "ope_no_liberal", "ope_new",
  "ope_reform", "ope_no_app_ecology", "ope_no_app_women", "ope_no_immigration",
  ↪ "ope_never_meaning",
  "agr_welfare", "agr_humanitarian", "agr_v_welfare", "agr_v_solidarity",
  ↪ "agr_v_compassion", "trust",
  "agr_no_give",
  "con_v_skills", "con_no_work", "con_imp_work", "con_success", "con_get",
  "sta_depressed", "sta_upset", "sta_control", "sta_serenity", "sta_no_meaning",
  "postmat",
  "social_class", "marital_status", "education", "education_levels", "employment_status",
  ↪
  "income_scale", "health",
  "religion"
)

# Reorder the columns in ts to match the defined order
ts <- ts[, column_order]

```

1.4 List relevant variables

The shortened database “ts” includes 224434 observations and 51 variables.

It consists of several categories of variables:

- **General/Demographic Variables:** Individual-level attributes such as `id` (respondent ID), `sex`, `age`, `weight` (sampling weight), `country_abb` (country of survey), and `survey_year` (year of survey).

- **Big Five Personality Traits** (candidate variables before a factor analysis):
 - **Extraversion:** `ext_excited` (ever felt very excited or interested), `ext_no_friends` (spend time with friends), `ext_no_colleagues` (spend time with colleagues from work), `ext_no_community` (spend time with people at sport, culture, communal organization), `ext_no_persuasion` (persuading friends, relatives or fellow workers), `ext_v_social` (reasons voluntary work: for social reasons), `ext_meeting` (important in a job: meeting people), `ext_convince` (personal characteristics: I enjoy convincing others of my opinion).
 - **Openness to Experience:** `ope_arts` (member: belong to education, arts, music or cultural activities), `ope_ecology` (member: belong to conservation, the environment, ecology, animal rights), `ope_v_arts` (voluntary work: unpaid work education, arts, music or cultural activities), `ope_v_ecology` (voluntary work: unpaid work environment, conservation, animal rights), `ope_no_liberal` (self positioning in political scale), `ope_new` (new and old ideas), `ope_reform` (political reform is moving too rapidly), `ope_no_app_ecology` (no approval: ecology movement or nature protection), `ope_no_app_women` (no approval: women's movement), `ope_no_immigration` (immigrant policy), `ope_never_meaning` (thinking about meaning and purpose of life).
 - **Agreeableness:** `agr_welfare` (member: belong to social welfare service for elderly, handicapped or deprived people), `agr_humanitarian` (member: belong to humanitarian or charitable organization), `agr_v_welfare` (voluntary work: unpaid work social welfare service for elderly, handicapped or deprived people), `agr_v_solidarity` (reasons voluntary work: solidarity with the poor and disadvantaged), `agr_v_compassion` (reasons voluntary work: compassion for those in need), `agr_trust` (do you think most people try to take advantage of you (10-point scale)), `agr_no_give` (would give part of my income for the environment).
 - **Conscientiousness:** `con_v_skills` (reasons voluntary work: gain new skills and useful experience), `con_no_work` (Work should come first even if it means less spare time), `con_imp_work` (why people work: work most important in my life), `con_success` (personal characteristics: I usually count on being successful in everything I do), `con_get` (personal characteristics: I am good at getting what I want).
 - **Emotional Stability (Inverse of Neuroticism):** `sta_depressed` (ever felt depressed or very unhappy), `sta_upset` (ever felt upset because somebody criticized you), `sta_control` (how much freedom of choice and control), `sta_serenity` (personal characteristics: changes, worry or welcome possibility), `sta_no_meaning` (life has no meaning).
- **Postmaterialist Values:** `postmat` (post-materialist index 12-item).
- **Environmental and Phenotypic Factors:**
 - `social_class` (subjective social class),
 - `marital_status`,
 - `education` (education attainment),
 - `education_levels` (education attainment simplified),
 - `employment_status` (employment category),
 - `income_scale` (scale of incomes),
 - `health` (self-assessed health).
- **Potential Confound:** `religion` (religious denomination).

2 Data cleaning

I check whether the variables contain inappropriate values (like “-1” or “-99”) using the “table” function and the codebook to interpret values. If this is the case, I convert these values into “NA”.

```

#### Clean the data ####

# Put NAs instead of irrelevant numerical values
variables_to_clean <- c(
  "sex", "age", "weight",
  "ext_excited", "ext_no_friends", "ext_no_colleagues", "ext_no_community",
  "ext_no_persuasion", "ext_v_social", "ext_meeting", "ext_convince",
  "ope_arts", "ope_ecology", "ope_v_arts", "ope_v_ecology", "ope_no_liberal", "ope_new",
  "ope_reform", "ope_no_app_ecology", "ope_no_app_women", "ope_no_immigration",
  ↪ "ope_never_meaning",
  "agr_welfare", "agr_humanitarian", "agr_v_welfare", "agr_v_solidarity",
  ↪ "agr_v_compassion", "trust",
  "agr_no_give",
  "con_v_skills", "con_no_work", "con_imp_work", "con_success", "con_get",
  "sta_depressed", "sta_upset", "sta_control", "sta_serenity", "sta_no_meaning",
  "postmat",
  "social_class", "marital_status", "education", "education_levels", "employment_status",
  ↪
  "income_scale", "health",
  "religion"
)

for (var in variables_to_clean) {
  ts[[var]][ts[[var]] %in% c(-5, -4, -3, -2, -1)] <- NA
}

# Create a new 'country' variable based on 'country_abb'
ts$country <- recode(ts$country_abb,
  "AL" = "Albania", "AM" = "Armenia", "AT" = "Austria", "AZ" = "Azerbaijan",
  "BA" = "Bosnia and Herzegovina", "BE" = "Belgium", "BG" = "Bulgaria", "BY" =
  ↪ "Belarus",
  "CA" = "Canada", "CH" = "Switzerland", "CY" = "Cyprus", "CY-TCC" = "Cyprus-TCC",
  "CZ" = "Czech Republic", "DE" = "Germany", "DK" = "Denmark", "EE" = "Estonia",
  "ES" = "Spain", "FI" = "Finland", "FR" = "France", "GB-GBN" = "United Kingdom (GBN)",
  ↪
  "GB-NIR" = "United Kingdom (Northern Ireland)", "GE" = "Georgia", "GR" = "Greece",
  "HR" = "Croatia", "HU" = "Hungary", "IE" = "Ireland", "IS" = "Iceland", "IT" =
  ↪ "Italy",
  "LT" = "Lithuania", "LU" = "Luxembourg", "LV" = "Latvia", "MD" = "Moldova",
  "ME" = "Montenegro", "MK" = "North Macedonia", "MT" = "Malta", "NL" = "Netherlands",
  "NO" = "Norway", "PL" = "Poland", "PT" = "Portugal", "RO" = "Romania", "RS" =
  ↪ "Serbia",
  "RS-KM" = "Serbia-Kosovo", "RU" = "Russia", "SE" = "Sweden", "SI" = "Slovenia",
  "SK" = "Slovakia", "TR" = "Turkey", "UA" = "Ukraine", "US" = "United States"
)

# Remove 'country_abb' as it's no longer needed
ts$country_abb <- NULL

# Modify the sex variable (1=0, 2=1)
ts$sex <- ts$sex - 1

# Create a new 'partner' variable based on 'marital_status'
ts$partner <- ifelse(ts$marital_status %in% c(1, 2, 8), 1, 0)

```

```

ts$marital_status <- NULL

# Create a new 'unemployment' variable based on 'employment_status'
ts$unemployment <- ifelse(ts$employment_status %in% c(7), 1, 0)
ts$employment_status <- NULL

# Invert the social_class variable so a higher score reflects a higher class
ts$social_class <- 6 - ts$social_class

# Invert the health variable so a higher score reflects a better health
ts$health <- 6 - ts$health

# Recode the sta_no_meaning variable
ts$sta_no_meaning <- c(`1` = 3,
                      `2` = 1,
                      `3` = 2)[as.character(ts$sta_no_meaning)]

# Convert haven-labelled variables to appropriate R types
numerical_vars <- c(
  "sex", "age", "weight", "survey_year",
  "ext_excited", "ext_no_friends", "ext_no_colleagues", "ext_no_community",
  "ext_no_persuasion",
  "ext_v_social", "ext_meeting", "ext_convince",
  "ope_arts", "ope_ecology", "ope_v_arts", "ope_v_ecology", "ope_no_liberal", "ope_new",
  "ope_reform", "ope_no_app_ecology", "ope_no_app_women", "ope_no_immigration",
  ↪ "ope_never_meaning",
  "agr_welfare", "agr_humanitarian", "agr_v_welfare", "agr_v_solidarity",
  ↪ "agr_v_compassion", "trust",
  "agr_no_give",
  "con_v_skills", "con_no_work", "con_imp_work", "con_success", "con_get",
  "sta_depressed", "sta_upset", "sta_control", "sta_serenity", "sta_no_meaning",
  "postmat",
  "social_class", "partner", "education", "education_levels", "unemployment",
  "income_scale", "health",
  "religion"
)

ts <- ts %>%
  mutate(
    # Convert 'id' to character
    id = as.character(id),
    # Convert other variables to numeric
    across(all_of(numerical_vars), as.numeric)
  )

```

Below, here is a table with information on demographic variables.

```

### Create a summary table (PDF document) ###

# Create the summary table
summary_table <- data.frame(
  Variable = c("Sex", "Age", "Social Class", "Income Scale", "Partner", "Unemployment",
    ↪ "Health"),

```

```

Non_NA = c(
  sum(!is.na(ts$sex)),
  sum(!is.na(ts$age)),
  sum(!is.na(ts$social_class)),
  sum(!is.na(ts$income_scale)),
  sum(!is.na(ts$partner)),
  sum(!is.na(ts$unemployment)),
  sum(!is.na(ts$health))
),

Missing = c(
  sum(is.na(ts$sex)),
  sum(is.na(ts$age)),
  sum(is.na(ts$social_class)),
  sum(is.na(ts$income_scale)),
  sum(is.na(ts$partner)),
  sum(is.na(ts$unemployment)),
  sum(is.na(ts$health))
),

# For 'Sex', 1=Female, 0=Male => mean is proportion of females
Mean = c(
  round(mean(ts$sex, na.rm = TRUE), 2),
  round(mean(ts$age, na.rm = TRUE), 2),
  round(mean(ts$social_class, na.rm = TRUE), 2),
  round(mean(ts$income_scale, na.rm = TRUE), 2),
  round(mean(ts$partner, na.rm = TRUE), 2),
  round(mean(ts$unemployment, na.rm = TRUE), 2),
  round(mean(ts$health, na.rm = TRUE), 2)
),

SD = c(
  round(sd(ts$sex, na.rm = TRUE), 2),
  round(sd(ts$age, na.rm = TRUE), 2),
  round(sd(ts$social_class, na.rm = TRUE), 2),
  round(sd(ts$income_scale, na.rm = TRUE), 2),
  round(sd(ts$partner, na.rm = TRUE), 2),
  round(sd(ts$unemployment, na.rm = TRUE), 2),
  round(sd(ts$health, na.rm = TRUE), 2)
),

Min = c(
  round(min(ts$sex, na.rm = TRUE), 2),
  round(min(ts$age, na.rm = TRUE), 2),
  round(min(ts$social_class, na.rm = TRUE), 2),
  round(min(ts$income_scale, na.rm = TRUE), 2),
  round(min(ts$partner, na.rm = TRUE), 2),
  round(min(ts$unemployment, na.rm = TRUE), 2),
  round(min(ts$health, na.rm = TRUE), 2)
),

Max = c(
  round(max(ts$sex, na.rm = TRUE), 2),
  round(max(ts$age, na.rm = TRUE), 2),

```

```

    round(max(ts$social_class, na.rm = TRUE), 2),
    round(max(ts$income_scale, na.rm = TRUE), 2),
    round(max(ts$partner, na.rm = TRUE), 2),
    round(max(ts$unemployment, na.rm = TRUE), 2),
    round(max(ts$health, na.rm = TRUE), 2)
  )
)

# Display the table as LaTeX (for inclusion in a PDF document)
kable(summary_table,
      format = "latex",
      booktabs = TRUE,
      caption = "Summary of Relevant Variables",
      label = "tab:summary") %>%
kable_styling(latex_options = c("hold_position"), font_size = 8)

```

Table 1: Summary of Relevant Variables

Variable	Non_NA	Missing	Mean	SD	Min	Max
Sex	224311	123	0.55	0.50	0	1
Age	223506	928	46.21	17.68	15	108
Social Class	843	223591	2.96	0.95	1	5
Income Scale	187462	36972	4.96	2.53	1	11
Partner	224434	0	0.59	0.49	0	1
Unemployment	224434	0	0.08	0.27	0	1
Health	182207	42227	3.73	0.95	1	5

3 Create outcome measures

In what follows, I create Big Five personality scores based on items in the European Values Study.

```

#### Psychometric analyses ####

# A) Safely compute alpha with suppressed messages/warnings
safe_alpha <- function(df) {
  # If <2 cols or <2 rows, alpha is meaningless
  if (ncol(df) < 2 || nrow(df) < 2) {
    return(NA_real_)
  }
  out <- tryCatch(
    suppressWarnings(
      suppressMessages(
        psych::alpha(df, check.keys = FALSE)$total$raw_alpha
      )
    ),
    error = function(e) NA_real_
  )
  return(out)
}

# B) Build personality scale with standardized scoring

```



```

build_personality_scale <- function(
  data,
  items,
  negative_items = NULL,
  scale_name = "some_trait",
  alpha_threshold = 0.50
) {
  cat("---- Building scale for", toupper(scale_name), "----\n")

  # Subset relevant items
  df <- data %>% select(all_of(items))

  # Reverse-code negative items
  if (!is.null(negative_items) && length(negative_items) > 0) {
    df <- df %>%
      mutate(across(all_of(negative_items), ~ {
        if (all(is.na(.))) {
          .
        } else {
          # Reverse code: newValue = max + min - oldValue
          v_max <- max(., na.rm = TRUE)
          v_min <- min(., na.rm = TRUE)
          v_max + v_min - .
        }
      }))
  }

  # Check alpha on all subsets of size >= 2
  if (length(items) >= 2) {
    all_subsets <- unlist(
      lapply(2:length(items), function(k) combn(items, k, simplify = FALSE)),
      recursive = FALSE
    )
    alpha_results <- map_df(all_subsets, function(subset_items) {
      subset_data <- df %>% select(all_of(subset_items)) %>% na.omit()
      this_alpha <- safe_alpha(subset_data)
      data.frame(
        Items = paste(subset_items, collapse = ", "),
        Num_Items = length(subset_items),
        CronAlpha = this_alpha
      )
    })
    max_alpha <- max(alpha_results$CronAlpha, na.rm = TRUE)
  } else {
    alpha_results <- data.frame(Items = NA, Num_Items = 1, CronAlpha = NA_real_)
    max_alpha <- NA_real_
  }

  cat("Highest alpha found for", scale_name, "=", round(max_alpha, 3), "\n")

  # Use multi-item or fallback to single
  if (is.na(max_alpha) || max_alpha < alpha_threshold) {
    cat(" -> No subset with alpha >=", alpha_threshold, ". Using single best item.\n")
    non_missing_counts <- sapply(df, function(x) sum(!is.na(x)))
  }
}

```

```

best_item <- names(non_missing_counts)[which.max(non_missing_counts)]

single_vec <- df[[best_item]]
if (all(is.na(single_vec))) {
  final_scale <- single_vec
} else {
  final_scale <- as.numeric(scale(single_vec, center = TRUE, scale = TRUE))
}
data[[paste0(scale_name, "_scale")]] <- final_scale

cat("    -> Created", paste0(scale_name, "_scale"),
    "based on standardized item:", best_item, "\n")
} else {
  best_row <- alpha_results[which.max(alpha_results$CronAlpha), ]
  chosen_items <- strsplit(as.character(best_row$Items), ", ")[[1]]

  cat(" -> Found subset with alpha =", round(best_row$CronAlpha, 3), "\n")
  cat("    Items:", best_row$Items, "\n")

  subset_data <- df[, chosen_items, drop = FALSE]
  scaled_mat <- scale(subset_data, center = TRUE, scale = TRUE)

  rowwise_means <- apply(scaled_mat, 1, function(x) {
    if (any(is.na(x))) NA_real_ else mean(x)
  })

  data[[paste0(scale_name, "_scale")]] <- rowwise_means
  cat("    -> Created", paste0(scale_name, "_scale"),
    "by averaging standardized chosen items.\n")
}

cat("---- End of scale building for", toupper(scale_name), "----\n\n")
return(data)
}

alpha_cutoff <- 0.50 #This is a low cutoff, 0.70 or 0.80 is reasonable usually

### Extraversion ###
extraversion_items <- c(
  "ext_excited", "ext_no_friends", "ext_no_colleagues", "ext_no_community",
  "ext_no_persuasion", "ext_v_social", "ext_meeting", "ext_convince"
)
extraversion_negative <- c(
  "ext_no_friends", "ext_no_colleagues", "ext_no_community", "ext_no_persuasion"
)
ts <- build_personality_scale(
  data = ts,
  items = extraversion_items,
  negative_items = extraversion_negative,
  scale_name = "extraversion",
  alpha_threshold = alpha_cutoff
)

```

```
## ---- Building scale for EXTRAVERSION ----
```

```
## Highest alpha found for extraversion = 0.521
## -> Found subset with alpha = 0.521
##     Items: ext_no_friends, ext_no_colleagues, ext_no_community
##     -> Created extraversion_scale by averaging standardized chosen items.
## ---- End of scale building for EXTRAVERSION ----
```

```
### Openness ###
final_openness_items <- c(
  "ope_arts", "ope_v_arts", "ope_no_liberal",
  "ope_no_app_ecology", "ope_no_app_women",
  "ope_no_immigration", "ope_never_meaning"
)
final_openness_negative <- c(
  "ope_no_liberal", "ope_no_app_ecology",
  "ope_no_app_women", "ope_no_immigration",
  "ope_never_meaning"
)
ts <- build_personality_scale(
  data = ts,
  items = final_openness_items,
  negative_items = final_openness_negative,
  scale_name = "openness",
  alpha_threshold = alpha_cutoff
)
```

```
## ---- Building scale for OPENNESS ----
## Some items ( ope_no_liberal ) were negatively correlated with the first principal component and
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' optionSome items ( ope_no_liberal ) we
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' optionSome items ( ope_no_liberal ) we
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' optionHighest alpha found for openness
## -> Found subset with alpha = 0.72
##     Items: ope_arts, ope_v_arts
##     -> Created openness_scale by averaging standardized chosen items.
## ---- End of scale building for OPENNESS ----
```

```
### Agreeableness ###
agreeableness_items <- c(
  "agr_welfare", "agr_humanitarian", "agr_v_welfare",
  "agr_v_solidarity", "agr_v_compassion",
  "trust",
  "agr_no_give"
)
agreeableness_negative <- c("agr_no_give")
ts <- build_personality_scale(
  data = ts,
  items = agreeableness_items,
  negative_items = agreeableness_negative,
  scale_name = "agreeableness",
  alpha_threshold = alpha_cutoff
)
```

```
## ---- Building scale for AGREEABLENESS ----
## Some items ( trust ) were negatively correlated with the first principal component and
## probably should be reversed.
## To do this, run the function again with the 'check.keys=TRUE' optionHighest alpha found for agreeabl
## -> Found subset with alpha = 0.797
##     Items: agr_v_solidarity, agr_v_compassion
##     -> Created agreeableness_scale by averaging standardized chosen items.
## ---- End of scale building for AGREEABLENESS ----
```

```
### Conscientiousness ###
conscientiousness_items <- c(
  "con_v_skills", "con_no_work", "con_imp_work", "con_success", "con_get"
)
conscientiousness_negative <- c("con_no_work")
ts <- build_personality_scale(
  data = ts,
  items = conscientiousness_items,
  negative_items = conscientiousness_negative,
  scale_name = "conscientiousness",
  alpha_threshold = alpha_cutoff
)
```

```
## ---- Building scale for CONSCIENTIOUSNESS ----
## Highest alpha found for conscientiousness = 0.351
## -> No subset with alpha >= 0.5 . Using single best item.
##     -> Created conscientiousness_scale based on standardized item: con_no_work
## ---- End of scale building for CONSCIENTIOUSNESS ----
```

```
### Emotional Stability ###
emotional_stability_items <- c(
  "sta_depressed", "sta_upset",
  "sta_control", "sta_serenity",
  "sta_no_meaning"
)
emotional_stability_negative <- c("sta_depressed", "sta_upset", "sta_no_meaning")
ts <- build_personality_scale(
  data = ts,
  items = emotional_stability_items,
  negative_items = emotional_stability_negative,
  scale_name = "emotional_stability",
  alpha_threshold = alpha_cutoff
)
```

```
## ---- Building scale for EMOTIONAL_STABILITY ----
## Highest alpha found for emotional_stability = 0.366
## -> No subset with alpha >= 0.5 . Using single best item.
##     -> Created emotional_stability_scale based on standardized item: sta_control
## ---- End of scale building for EMOTIONAL_STABILITY ----
```

```
### Remove useless variables ###

# Combine all item names into a single vector
```

```

all_scale_items <- c(
  # Extraversion
  "ext_excited", "ext_no_friends", "ext_no_colleagues", "ext_no_community",
  "ext_no_persuasion", "ext_v_social", "ext_meeting", "ext_convince",

  # Openness
  "ope_arts", "ope_ecology", "ope_v_arts", "ope_v_ecology",
  "ope_no_liberal", "ope_new", "ope_reform",
  "ope_no_app_ecology", "ope_no_app_women", "ope_no_immigration",
  "ope_never_meaning",

  # Agreeableness
  "agr_welfare", "agr_humanitarian", "agr_v_welfare", "agr_v_solidarity",
  "agr_v_compassion", "trust", "agr_no_give",

  # Conscientiousness
  "con_v_skills", "con_no_work", "con_imp_work", "con_success", "con_get",

  # Emotional Stability
  "sta_depressed", "sta_upset", "sta_control", "sta_serenity", "sta_no_meaning"
)

# Remove the items
ts <- ts %>%
  select(-dplyr::any_of(all_scale_items))

```

4 Main statistical analyses

In the following part, we look at the relationship between socio-economic status and personality traits.

```

### Standard linear regressions controlling for sex and age ###
ts$age_z <- scale(ts$age, center=TRUE, scale=TRUE)
ts$income_scale_z <- scale(ts$income_scale, center=TRUE, scale=TRUE)
ts$education_z <- scale(ts$education, center=TRUE, scale=TRUE)

extraversion_lm <- lm(
  extraversion_scale ~ income_scale_z + sex + age_z,
  data = ts,
  weights = weight
)
summary(extraversion_lm)

```

```

##
## Call:
## lm(formula = extraversion_scale ~ income_scale_z + sex + age_z,
##     data = ts, weights = weight)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -3.8848 -0.4513  0.0090  0.4561  2.7681
##

```

```
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.039246   0.006023   6.516 7.37e-11 ***
## income_scale_z 0.084478   0.004262  19.823 < 2e-16 ***
## sex          -0.177762   0.008507 -20.895 < 2e-16 ***
## age_z        -0.247633   0.004779 -51.816 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.66 on 24142 degrees of freedom
## (200288 observations effacées parce que manquantes)
## Multiple R-squared:  0.15, Adjusted R-squared:  0.1499
## F-statistic: 1421 on 3 and 24142 DF, p-value: < 2.2e-16
```

```
openness_lm <- lm(
  openness_scale ~ income_scale_z + sex + age_z,
  data = ts,
  weights = weight
)
summary(openness_lm)
```

```
##
## Call:
## lm(formula = openness_scale ~ income_scale_z + sex + age_z, data = ts,
##     weights = weight)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -0.9353 -0.3082 -0.2290 -0.1538  7.9493
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.022627   0.003454  -6.551 5.74e-11 ***
## income_scale_z  0.084732   0.002520  33.622 < 2e-16 ***
## sex           0.014786   0.004809   3.075 0.00211 **
## age_z        -0.018590   0.002516  -7.388 1.50e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8706 on 132663 degrees of freedom
## (91765 observations effacées parce que manquantes)
## Multiple R-squared:  0.01016, Adjusted R-squared:  0.01014
## F-statistic: 454 on 3 and 132663 DF, p-value: < 2.2e-16
```

```
agreeableness_lm <- lm(
  agreeableness_scale ~ income_scale_z + sex + age_z,
  data = ts,
  weights = weight
)
summary(agreeableness_lm)
```

```
##
```

```
## Call:
## lm(formula = agreeableness_scale ~ income_scale_z + sex + age_z,
##     data = ts, weights = weight)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8495 -0.5163  0.1110  0.6626  2.8482
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.06398   0.01348  -4.746 2.11e-06 ***
## income_scale_z -0.06401   0.01015  -6.308 2.98e-10 ***
## sex           0.22583   0.01947  11.601 < 2e-16 ***
## age_z         0.16731   0.01121  14.926 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8686 on 8157 degrees of freedom
## (216273 observations effacées parce que manquantes)
## Multiple R-squared:  0.05346,    Adjusted R-squared:  0.05311
## F-statistic: 153.6 on 3 and 8157 DF,  p-value: < 2.2e-16
```

```
conscientiousness_lm <- lm(
  conscientiousness_scale ~ income_scale_z + sex + age_z,
  data = ts,
  weights = weight
)
summary(conscientiousness_lm)
```

```
##
## Call:
## lm(formula = conscientiousness_scale ~ income_scale_z + sex +
##     age_z, data = ts, weights = weight)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -7.0694 -0.7897  0.0703  0.7170  5.6866
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.056616   0.003820  14.82  <2e-16 ***
## income_scale_z -0.118538   0.002696 -43.97  <2e-16 ***
## sex           -0.113788   0.005333 -21.34  <2e-16 ***
## age_z          0.186404   0.002742  67.99  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9756 on 135305 degrees of freedom
## (89122 observations effacées parce que manquantes)
## Multiple R-squared:  0.05693,    Adjusted R-squared:  0.05691
## F-statistic: 2723 on 3 and 135305 DF,  p-value: < 2.2e-16
```

```

emotional_stability_lm <- lm(
  emotional_stability_scale ~ income_scale_z + sex + age_z,
  data = ts,
  weights = weight
)
summary(emotional_stability_lm)

##
## Call:
## lm(formula = emotional_stability_scale ~ income_scale_z + sex +
##     age_z, data = ts, weights = weight)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -9.1549 -0.6305  0.1174  0.6615  5.3128
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.016395   0.003348   4.897 9.71e-07 ***
## income_scale_z 0.126025   0.002382  52.917 < 2e-16 ***
## sex          -0.049670   0.004679 -10.616 < 2e-16 ***
## age_z         -0.048641   0.002423 -20.074 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9949 on 182804 degrees of freedom
## (41623 observations effacées parce que manquantes)
## Multiple R-squared:  0.02169,    Adjusted R-squared:  0.02167
## F-statistic: 1351 on 3 and 182804 DF,  p-value: < 2.2e-16

```

Mixed-effects models controlling for sex and age with random intercepts for country
 ↳ *####*

```

extraversion_lmer <- lmer(
  extraversion_scale ~ income_scale_z + sex + age_z + (1 | country),
  data = ts,
  weights = weight
)
summary(extraversion_lmer)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: extraversion_scale ~ income_scale_z + sex + age_z + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: 47773.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.8256 -0.6836  0.0020  0.6880  4.0492
##

```



```
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   country (Intercept) 0.04215  0.2053
##   Residual          0.39387  0.6276
## Number of obs: 24146, groups:  country, 32
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    5.166e-02  3.678e-02  3.174e+01   1.405    0.17
## income_scale_z  6.327e-02  4.325e-03  2.414e+04  14.629 <2e-16 ***
## sex            -1.675e-01  8.139e-03  2.412e+04 -20.585 <2e-16 ***
## age_z          -2.475e-01  4.591e-03  2.413e+04 -53.912 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) incm__ sex
## incom_scl_z -0.006
## sex          -0.108  0.060
## age_z         0.021  0.206 -0.019
```

```
openness_lmer <- lmer(
  openness_scale ~ income_scale_z + sex + age_z + (1 | country),
  data      = ts,
  weights = weight
)
summary(openness_lmer)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: openness_scale ~ income_scale_z + sex + age_z + (1 | country)
##   Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.0194 -0.3728 -0.2281 -0.0999  9.1950
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   country (Intercept) 0.02572  0.1604
##   Residual          0.73503  0.8573
## Number of obs: 132669, groups:  country, 48
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -4.103e-02  2.344e-02  3.870e+01  -1.750 0.087987 .
## income_scale_z  6.611e-02  2.591e-03  1.324e+05  25.518 < 2e-16 ***
## sex            1.827e-02  4.739e-03  1.326e+05   3.856 0.000116 ***
## age_z         -2.341e-02  2.499e-03  1.327e+05  -9.370 < 2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) incm__ sex
## incom_scl_z -0.007
## sex          -0.105  0.082
## age_z         0.011  0.234 -0.029
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```
agreeableness_lmer <- lmer(
  agreeableness_scale ~ income_scale_z + sex + age_z + (1 | country),
  data      = ts,
  weights = weight
)
summary(agreeableness_lmer)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: agreeableness_scale ~ income_scale_z + sex + age_z + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: 21305.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4610 -0.6358  0.0987  0.7338  3.1792
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## country (Intercept) 0.06783  0.2604
## Residual              0.69553  0.8340
## Number of obs: 8161, groups: country, 27
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.06974    0.05213   28.02848  -1.338    0.192
## income_scale_z -0.06196    0.01058  8116.08736  -5.854 4.97e-09 ***
## sex           0.19839    0.01881  8139.28040  10.546 < 2e-16 ***
## age_z         0.15766    0.01090  8145.39978  14.464 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) incm__ sex
## incom_scl_z -0.047
## sex          -0.172  0.084
## age_z         0.043  0.165 -0.030
```

```
conscientiousness_lmer <- lmer(
  conscientiousness_scale ~ income_scale_z + sex + age_z + (1 | country),
  data      = ts,
```

```

    weights = weight
  )
summary(conscientiousness_lmer)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: conscientiousness_scale ~ income_scale_z + sex + age_z + (1 |
##   country)
##   Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##   Min       1Q   Median       3Q      Max
## -7.3737 -0.7225  0.0340  0.7309  6.7362
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
##   country  (Intercept)  0.09735   0.3120
##   Residual                    0.84968   0.9218
## Number of obs: 135312, groups:  country, 47
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    9.071e-02  4.568e-02  5.805e+01   1.986   0.0518 .
## income_scale_z -6.832e-02  2.647e-03  1.353e+05 -25.813 <2e-16 ***
## sex            -1.098e-01  5.044e-03  1.353e+05 -21.765 <2e-16 ***
## age_z           2.169e-01  2.624e-03  1.353e+05  82.676 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) incm__ sex
## incom_scl_z -0.002
## sex          -0.057  0.078
## age_z         0.002  0.201 -0.027
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs

emotional_stability_lmer <- lmer(
  emotional_stability_scale ~ income_scale_z + sex + age_z + (1 | country),
  data = ts,
  weights = weight
)
summary(emotional_stability_lmer)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emotional_stability_scale ~ income_scale_z + sex + age_z + (1 |
##   country)
##   Data: ts

```

```
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.3871 -0.6057  0.1086  0.6729  6.4100
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   country  (Intercept) 0.04385  0.2094
##   Residual              0.95336  0.9764
## Number of obs: 182811, groups:  country, 49
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   2.348e-02  3.013e-02  4.164e+01   0.779    0.44
## income_scale_z  9.951e-02  2.424e-03  1.827e+05  41.054 <2e-16 ***
## sex           -4.769e-02  4.595e-03  1.828e+05 -10.378 <2e-16 ***
## age_z         -5.191e-02  2.399e-03  1.828e+05 -21.640 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) incm__ sex
## incom_scl_z -0.005
## sex         -0.078  0.082
## age_z        0.006  0.212 -0.023
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

5 Robustness checks

I first check several hypotheses underlying these analyses.

I then conduct extra tests to evaluate this relationship between income and personality replicates under slightly different conditions. Specifically, I observe whether education, which is also an indicator of socioeconomic status produces similar results. I also evaluate whether religion constitutes a confounding variable in these analyses.

```
### Examine the normality of the residuals ###

# Extract raw conditional residuals
resid_extraversion <- residuals(extraversion_lmer)
resid_openness <- residuals(openness_lmer)
resid_agreeableness <- residuals(agreeableness_lmer)
resid_conscientiousness <- residuals(conscientiousness_lmer)
resid_emotional_stability <- residuals(emotional_stability_lmer)

# QQ-plot
qqnorm(resid_extraversion, main = "QQ Plot of Extraversion Residuals");
↪ qqline(resid_extraversion, col = "red")
```

```
#qqnorm(resid_openness, main = "QQ Plot of Openness Residuals"); qqline(resid_openness,
↪ col = "red")

#qqnorm(resid_agreeableness, main = "QQ Plot of Agreeableness Residuals");
↪ qqline(resid_agreeableness, col = "red")

#qqnorm(resid_conscientiousness, main = "QQ Plot of Conscientiousness Residuals");
↪ qqline(resid_conscientiousness, col = "red")

#qqnorm(resid_emotional_stability, main = "QQ Plot of Emotional Stability Residuals");
↪ qqline(resid_emotional_stability, col = "red")

### Test for collinearity ###

check_collinearity(extraversion_lmer)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##          Term  VIF    VIF 95% CI Increased SE Tolerance Tolerance 95% CI
## income_scale_z 1.05 [1.04, 1.06]         1.02      0.95      [0.94, 0.96]
##          sex 1.00 [1.00, 1.07]         1.00      1.00      [0.93, 1.00]
##          age_z 1.05 [1.03, 1.06]         1.02      0.96      [0.94, 0.97]
```

```
check_collinearity(openness_lmer)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##          Term  VIF    VIF 95% CI Increased SE Tolerance Tolerance 95% CI
## income_scale_z 1.07 [1.06, 1.07]         1.03      0.94      [0.93, 0.94]
##          sex 1.01 [1.01, 1.02]         1.00      0.99      [0.98, 0.99]
##          age_z 1.06 [1.05, 1.07]         1.03      0.94      [0.94, 0.95]
```

```
check_collinearity(agreeableness_lmer)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##          Term  VIF    VIF 95% CI Increased SE Tolerance Tolerance 95% CI
## income_scale_z 1.04 [1.02, 1.07]         1.02      0.96      [0.94, 0.98]
##          sex 1.01 [1.00, 1.10]         1.00      0.99      [0.91, 1.00]
##          age_z 1.03 [1.01, 1.06]         1.01      0.97      [0.94, 0.99]
```

```
check_collinearity(conscientiousness_lmer)
```

```
## # Check for Multicollinearity
```

```
##
## Low Correlation
##
##          Term   VIF   VIF 95% CI Increased SE Tolerance Tolerance 95% CI
## income_scale_z 1.05 [1.04, 1.06]         1.02      0.95      [0.95, 0.96]
##          sex 1.01 [1.00, 1.02]         1.00      0.99      [0.98, 1.00]
##          age_z 1.04 [1.04, 1.05]         1.02      0.96      [0.95, 0.96]
```

```
check_collinearity(emotional_stability_lmer)
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##          Term   VIF   VIF 95% CI Increased SE Tolerance Tolerance 95% CI
## income_scale_z 1.06 [1.05, 1.06]         1.03      0.95      [0.94, 0.95]
##          sex 1.01 [1.00, 1.01]         1.00      0.99      [0.99, 1.00]
##          age_z 1.05 [1.04, 1.05]         1.02      0.95      [0.95, 0.96]
```

```
### Mixed-effects models with an interaction effect between income and age ###
```

```
extraversion_lmer_interaction <- lmer(
  extraversion_scale ~ income_scale_z*age_z + sex + (1 | country),
  data = ts,
  weights = weight
)
summary(extraversion_lmer_interaction)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: extraversion_scale ~ income_scale_z * age_z + sex + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: 47776.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.8147 -0.6819  0.0011  0.6890  4.0340
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.04239 0.2059
## Residual 0.39378 0.6275
## Number of obs: 24146, groups: country, 32
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  4.975e-02  3.689e-02 3.176e+01  1.349    0.1870
## income_scale_z  6.144e-02  4.385e-03 2.414e+04 14.011 <2e-16 ***
## age_z        -2.503e-01  4.722e-03 2.412e+04 -53.005 <2e-16 ***
## sex          -1.689e-01  8.156e-03 2.412e+04 -20.708 <2e-16 ***
```

```

## income_scale_z:age_z -1.137e-02  4.538e-03  2.412e+04  -2.506   0.0122 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) incm__ age_z  sex
## incom_scl_z -0.003
## age_z       0.025  0.236
## sex        -0.106  0.070 -0.003
## incm_scl:_  0.021  0.166  0.234  0.066

openness_lmer_interaction <- lmer(
  openness_scale ~ income_scale_z*age_z + sex + (1 | country),
  data = ts,
  weights = weight
)
summary(openness_lmer_interaction)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: openness_scale ~ income_scale_z * age_z + sex + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.8325 -0.3732 -0.2313 -0.1025  9.2251
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.02569 0.1603
## Residual 0.73413 0.8568
## Number of obs: 132669, groups: country, 48
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -3.543e-02  2.343e-02 3.801e+01  -1.512    0.139
## income_scale_z    6.978e-02  2.605e-03 1.324e+05  26.788 < 2e-16 ***
## age_z         -1.612e-02  2.562e-03 1.327e+05  -6.293 3.13e-10 ***
## sex           2.134e-02  4.742e-03 1.326e+05   4.500 6.81e-06 ***
## income_scale_z:age_z 3.293e-02  2.579e-03 1.326e+05  12.771 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) incm__ age_z  sex
## incom_scl_z -0.005
## age_z       0.015  0.251
## sex        -0.103  0.087 -0.017
## incm_scl:_  0.019  0.110  0.223  0.051
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs

```

```
agreeableness_lmer_interaction <- lmer(
  agreeableness_scale ~ income_scale_z*age_z + sex + (1 | country),
  data = ts,
  weights = weight
)
summary(agreeableness_lmer_interaction)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: agreeableness_scale ~ income_scale_z * age_z + sex + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: 21312
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.4601 -0.6366  0.0984  0.7340  3.1769
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.06796 0.2607
## Residual 0.69559 0.8340
## Number of obs: 8161, groups: country, 27
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -6.914e-02  5.219e-02 2.805e+01  -1.325    0.196
## income_scale_z  -6.102e-02  1.074e-02 8.128e+03  -5.680 1.39e-08 ***
## age_z          1.583e-01  1.097e-02 8.144e+03  14.428 < 2e-16 ***
## sex            1.989e-01  1.884e-02 8.138e+03  10.558 < 2e-16 ***
## income_scale_z:age_z  5.797e-03  1.119e-02 8.144e+03   0.518    0.604
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) incm__ age_z sex
## incom_scl_z -0.042
## age_z        0.045 0.181
## sex          -0.170 0.092 -0.024
## incm_scl_:_  0.022 0.170 0.114 0.052
```

```
conscientiousness_lmer_interaction <- lmer(
  conscientiousness_scale ~ income_scale_z*age_z + sex + (1 | country),
  data = ts,
  weights = weight
)
summary(conscientiousness_lmer_interaction)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: conscientiousness_scale ~ income_scale_z * age_z + sex + (1 |
```



```
##      country)
##      Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.3886 -0.7227  0.0331  0.7299  6.6577
##
## Random effects:
##      Groups   Name      Variance Std.Dev.
##  country   (Intercept) 0.09732  0.3120
##  Residual                0.84943  0.9216
## Number of obs: 135312, groups:  country, 47
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    9.300e-02  4.567e-02 5.823e+01  2.036   0.0463 *
## income_scale_z  -6.739e-02  2.650e-03 1.353e+05 -25.425 < 2e-16 ***
## age_z           2.213e-01  2.713e-03 1.353e+05  81.590 < 2e-16 ***
## sex            -1.081e-01  5.050e-03 1.353e+05 -21.397 < 2e-16 ***
## income_scale_z:age_z 1.710e-02  2.682e-03 1.353e+05   6.375 1.83e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) incm__ age_z  sex
## incom_scl_z -0.001
## age_z        0.004  0.209
## sex          -0.056  0.081 -0.012
## incm_scl_:_  0.008  0.055  0.254  0.053
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```
emotional_stability_lmer_interaction <- lmer(
  emotional_stability_scale ~ income_scale_z*age_z + sex + (1 | country),
  data      = ts,
  weights = weight
)
summary(emotional_stability_lmer_interaction)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emotional_stability_scale ~ income_scale_z * age_z + sex + (1 |
##      country)
##      Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -8.3892 -0.6062 0.1087 0.6732 6.4046
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.04385 0.2094
## Residual 0.95335 0.9764
## Number of obs: 182811, groups: country, 49
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 2.401e-02 3.013e-02 4.164e+01 0.797 0.43
## income_scale_z 9.977e-02 2.431e-03 1.827e+05 41.043 <2e-16 ***
## age_z -5.115e-02 2.459e-03 1.828e+05 -20.806 <2e-16 ***
## sex -4.737e-02 4.601e-03 1.828e+05 -10.295 <2e-16 ***
## income_scale_z:age_z 3.410e-03 2.429e-03 1.828e+05 1.404 0.16
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) incm__ age_z sex
## incom_scl_z -0.004
## age_z 0.008 0.223
## sex -0.078 0.085 -0.012
## incm_scl_:_ 0.013 0.075 0.219 0.050
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

Mixed-effects models with education instead of income

```
extraversion_lmer_education <- lmer(
  extraversion_scale ~ education_z + sex + age_z + (1 | country),
  data = ts,
  weights = weight
)
summary(extraversion_lmer_education)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: extraversion_scale ~ education_z + sex + age_z + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: 57234.3
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -5.6888 -0.6883 -0.0078 0.6768 4.3944
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.04705 0.2169
## Residual 0.39350 0.6273
## Number of obs: 28868, groups: country, 33
```

```
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  8.968e-02  3.816e-02  3.262e+01    2.35   0.025 *
## education_z  9.259e-02  3.812e-03  2.886e+04   24.29  <2e-16 ***
## sex         -1.919e-01  7.407e-03  2.884e+04  -25.91  <2e-16 ***
## age_z       -2.354e-01  4.187e-03  2.884e+04  -56.23  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) edctn_ sex
## education_z  0.027
## sex         -0.096 -0.002
## age_z       0.028  0.268 -0.031

openness_lmer_education <- lmer(
  openness_scale ~ education_z + sex + age_z + (1 | country),
  data = ts,
  weights = weight
)
summary(openness_lmer_education)

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: openness_scale ~ education_z + sex + age_z + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.3361 -0.3897 -0.2136 -0.0356  8.7375
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.03922 0.1981
## Residual 0.71862 0.8477
## Number of obs: 101019, groups: country, 46
##
## Fixed effects:
##           Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) -8.579e-03  2.948e-02  5.536e+01  -0.291 0.772137
## education_z  1.277e-01  2.871e-03  1.010e+05  44.502 < 2e-16 ***
## sex         2.201e-02  5.353e-03  1.010e+05   4.112 3.93e-05 ***
## age_z       -9.475e-03  2.849e-03  1.010e+05  -3.326 0.000882 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) edctn_ sex
## education_z  0.011
```

```
## sex          -0.095  0.029
## age_z        0.014  0.294 -0.046
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```
#agreeableness_lmer_education <- lmer(
# agreeableness_scale ~ scale(education_levels) + sex + age_z + (1 | country),
# data      = ts,
# weights = weight
#)
#summary(agreeableness_lmer_education)
#NB: 0 non-NA observations with agreeableness

conscientiousness_lmer_education <- lmer(
  conscientiousness_scale ~ education_z + sex + age_z + (1 | country),
  data      = ts,
  weights = weight
)
summary(conscientiousness_lmer_education)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: conscientiousness_scale ~ education_z + sex + age_z + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.8236 -0.7149  0.0433  0.7327  6.4056
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.087 0.2950
## Residual 0.846 0.9198
## Number of obs: 160691, groups: country, 47
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 7.296e-02 4.317e-02 6.739e+01 1.69 0.0956 .
## education_z -1.122e-01 2.489e-03 1.607e+05 -45.10 <2e-16 ***
## sex -1.019e-01 4.598e-03 1.606e+05 -22.17 <2e-16 ***
## age_z 1.981e-01 2.411e-03 1.607e+05 82.19 <2e-16 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) edctn_ sex
## education_z 0.004
## sex -0.055 0.019
## age_z 0.005 0.265 -0.040
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```

emotional_stability_lmer_education <- lmer(
  emotional_stability_scale ~ education_z + sex + age_z + (1 | country),
  data = ts,
  weights = weight
)
summary(emotional_stability_lmer_education)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emotional_stability_scale ~ education_z + sex + age_z + (1 |
##      country)
##      Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -9.3784 -0.5989  0.0991  0.6624  6.7632
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
##      country (Intercept) 0.04582  0.2141
##      Residual              0.96044  0.9800
## Number of obs: 161218, groups:  country, 47
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.050e-02  3.145e-02  5.196e+01  1.924   0.0599 .
## education_z  1.026e-01  2.640e-03  1.612e+05 38.872 <2e-16 ***
## sex         -5.985e-02  4.891e-03  1.612e+05 -12.237 <2e-16 ***
## age_z       -5.357e-02  2.565e-03  1.612e+05 -20.883 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) edctn_ sex
## education_z  0.006
## sex         -0.081  0.018
## age_z       0.008  0.262 -0.040
## optimizer (nlptwrap) convergence code: 0 (OK)
## Gradient contains NAs

```

Mixed-effects models controlling for the effect of religion

```

extraversion_lmer_religion <- lmer(
  extraversion_scale ~ income_scale_z + sex + age_z + religion + (1 | country),
  data = ts,
  weights = weight
)
summary(extraversion_lmer_religion)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [

```

```
## lmerModLmerTest]
## Formula: extraversion_scale ~ income_scale_z + sex + age_z + religion +
##      (1 | country)
##      Data: ts
## Weights: weight
##
## REML criterion at convergence: 47722
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.8342 -0.6826  0.0012  0.6894  4.0460
##
## Random effects:
##      Groups   Name      Variance Std.Dev.
## country (Intercept) 0.04213  0.2053
## Residual          0.39395  0.6277
## Number of obs: 24114, groups: country, 32
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   4.493e-02  3.746e-02  3.417e+01   1.199   0.239
## income_scale_z 6.332e-02  4.328e-03  2.410e+04  14.628 <2e-16 ***
## sex           -1.682e-01  8.181e-03  2.409e+04 -20.562 <2e-16 ***
## age_z         -2.479e-01  4.629e-03  2.410e+04 -53.546 <2e-16 ***
## religion       9.658e-03  1.004e-02  2.405e+04   0.962   0.336
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) incm__ sex    age_z
## incom_scl_z -0.010
## sex         -0.088  0.058
## age_z        0.044  0.202 -0.007
## religion    -0.191  0.022 -0.092 -0.122
```

```
openness_lmer_religion <- lmer(
  openness_scale ~ income_scale_z + sex + age_z + religion + (1 | country),
  data      = ts,
  weights = weight
)
summary(openness_lmer_religion)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: openness_scale ~ income_scale_z + sex + age_z + religion + (1 |
##      country)
##      Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -2.0108 -0.3725 -0.2281 -0.1002  9.1979
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   country (Intercept) 0.02576  0.1605
##   Residual              0.73506  0.8574
## Number of obs: 132155, groups:  country, 48
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -3.436e-02  2.390e-02  4.137e+01  -1.438    0.158
## income_scale_z  6.591e-02  2.596e-03  1.318e+05  25.388 < 2e-16 ***
## sex           1.865e-02  4.764e-03  1.321e+05   3.914 9.07e-05 ***
## age_z        -2.320e-02  2.524e-03  1.321e+05  -9.195 < 2e-16 ***
## religion      -8.588e-03  6.077e-03  1.308e+05  -1.413    0.158
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) incm__ sex    age_z
## incom_scl_z -0.007
## sex          -0.087  0.082
## age_z         0.035  0.233 -0.018
## religion     -0.191 -0.003 -0.079 -0.125
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```
agreeableness_lmer_religion <- lmer(
  agreeableness_scale ~ income_scale_z + sex + age_z + religion + (1 | country),
  data      = ts,
  weights = weight
)
summary(agreeableness_lmer_religion)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: agreeableness_scale ~ income_scale_z + sex + age_z + religion +
##          (1 | country)
##   Data: ts
## Weights: weight
##
## REML criterion at convergence: 21249.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5033 -0.6377  0.0993  0.7302  3.1111
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   country (Intercept) 0.06854  0.2618
##   Residual              0.69328  0.8326
## Number of obs: 8148, groups:  country, 27
##
```

```
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -0.15986    0.05505   34.02244  -2.904  0.00643 **
## income_scale_z -0.06108    0.01058  8101.29417  -5.775  7.96e-09 ***
## sex           0.19141    0.01884  8125.68437   10.162 < 2e-16 ***
## age_z         0.14961    0.01099  8130.23381   13.609 < 2e-16 ***
## religion      0.12619    0.02374  8025.27827    5.316 1.09e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) incm__ sex    age_z
## incom_scl_z -0.049
## sex         -0.142  0.083
## age_z       0.083  0.162 -0.020
## religion    -0.307  0.016 -0.066 -0.136
```

```
conscientiousness_lmer_religion <- lmer(
  conscientiousness_scale ~ income_scale_z + sex + age_z + religion + (1 | country),
  data = ts,
  weights = weight
)
summary(conscientiousness_lmer_religion)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: conscientiousness_scale ~ income_scale_z + sex + age_z + religion +
## (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.0713 -0.7208  0.0340  0.7301  6.5531
##
## Random effects:
## Groups   Name            Variance Std.Dev.
## country (Intercept) 0.09729  0.3119
## Residual              0.84651  0.9201
## Number of obs: 134686, groups: country, 47
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   -6.792e-03  4.589e-02  5.379e+01  -0.148    0.883
## income_scale_z -6.762e-02  2.648e-03  1.347e+05 -25.533 <2e-16 ***
## sex           -1.190e-01  5.064e-03  1.346e+05 -23.504 <2e-16 ***
## age_z         2.108e-01  2.641e-03  1.347e+05  79.815 <2e-16 ***
## religion      1.350e-01  6.255e-03  1.347e+05  21.591 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```



```
## Correlation of Fixed Effects:
##          (Intr) incm__ sex    age_z
## incom_scl_z -0.003
## sex         -0.048  0.077
## age_z        0.013  0.199 -0.017
## religion     -0.099  0.012 -0.083 -0.108
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```
emotional_stability_lmer_religion <- lmer(
  emotional_stability_scale ~ income_scale_z + sex + age_z + religion + (1 | country),
  data      = ts,
  weights = weight
)
summary(emotional_stability_lmer_religion)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emotional_stability_scale ~ income_scale_z + sex + age_z + religion +
##          (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -8.3581 -0.6063  0.1093  0.6731  6.3581
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.04389 0.2095
## Residual 0.95319 0.9763
## Number of obs: 182050, groups: country, 49
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  4.129e-02  3.045e-02  4.419e+01  1.356    0.182
## income_scale_z 9.912e-02  2.429e-03  1.820e+05 40.809 < 2e-16 ***
## sex          -4.639e-02  4.620e-03  1.820e+05 -10.041 < 2e-16 ***
## age_z         -5.084e-02  2.420e-03  1.820e+05 -21.010 < 2e-16 ***
## religion      -2.356e-02  5.766e-03  1.814e+05 -4.086 4.39e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) incm__ sex    age_z
## incom_scl_z -0.005
## sex         -0.066  0.082
## age_z        0.022  0.211 -0.014
## religion     -0.141 -0.001 -0.080 -0.114
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

6 Mediation analyses

In the following analyses, I first look at the relationship between health and personality traits, and then observe the extent to which health mediate the relationship between socio-economic status and personality.

```
### Mixed-effects models on the relationship between health and personality traits ###
```

```
#extraversion_lmer_health <- lmer(  
#  extraversion_scale ~ scale(health) + sex + age_z + (1 | country),  
#  data      = ts,  
#  weights = weight  
#)  
#summary(extraversion_lmer_health)  
#NB: 0 non-NA observations with extraversion
```

```
openness_lmer_health <- lmer(  
  openness_scale ~ scale(health) + sex + age_z + (1 | country),  
  data      = ts,  
  weights = weight  
)  
summary(openness_lmer_health)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [  
## lmerModLmerTest]  
## Formula: openness_scale ~ scale(health) + sex + age_z + (1 | country)  
## Data: ts  
## Weights: weight  
##  
## REML criterion at convergence: Inf  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max   
## -1.6600 -0.3585 -0.2370 -0.1161  8.4044   
##  
## Random effects:  
## Groups Name Variance Std.Dev.  
## country (Intercept) 0.02153 0.1467  
## Residual 0.72531 0.8517  
## Number of obs: 117319, groups: country, 48  
##  
## Fixed effects:  
##              Estimate Std. Error      df t value Pr(>|t|)      
## (Intercept) -4.834e-02  2.152e-02 3.999e+01 -2.246  0.03032 *   
## scale(health) 4.145e-02  2.807e-03 1.166e+05 14.769 < 2e-16 ***  
## sex          1.442e-02  5.000e-03 1.173e+05  2.884  0.00393 **  
## age_z        -2.186e-02  2.727e-03 1.173e+05 -8.015 1.11e-15 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Correlation of Fixed Effects:  
##              (Intr) scl(h) sex  
## scale(hlth) -0.003  
## sex         -0.121  0.073
```

```
## age_z          0.015  0.381 -0.025
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```
agreeableness_lmer_health <- lmer(
  agreeableness_scale ~ scale(health) + sex + age_z + (1 | country),
  data      = ts,
  weights = weight
)
summary(agreeableness_lmer_health)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: agreeableness_scale ~ scale(health) + sex + age_z + (1 | country)
## Data: ts
## Weights: weight
##
## REML criterion at convergence: 24765.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2542 -0.6576  0.0965  0.7370  3.1369
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.06734 0.2595
## Residual 0.72575 0.8519
## Number of obs: 9345, groups: country, 28
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  -0.09281    0.05091  28.93333  -1.823   0.0786 .
## scale(health) -0.01108    0.01049 9302.86256  -1.056   0.2909
## sex           0.20507    0.01788 9321.36959  11.470 <2e-16 ***
## age_z         0.16185    0.01048 9336.11918  15.439 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) scl(h) sex
## scale(hlth) -0.019
## sex         -0.168  0.054
## age_z       0.044  0.291 -0.033
```

```
conscientiousness_lmer_health <- lmer(
  conscientiousness_scale ~ scale(health) + sex + age_z + (1 | country),
  data      = ts,
  weights = weight
)
summary(conscientiousness_lmer_health)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
```

```
## lmerModLmerTest]
## Formula: conscientiousness_scale ~ scale(health) + sex + age_z + (1 |
##   country)
##   Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##   Min      1Q  Median      3Q      Max
## -7.1737 -0.7223  0.0292  0.7213  7.0929
##
## Random effects:
##   Groups   Name                Variance Std.Dev.
##   country  (Intercept)  0.09648   0.3106
##   Residual                    0.85322   0.9237
## Number of obs: 123030, groups:  country, 47
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)  6.907e-02  4.549e-02  6.262e+01  1.519   0.134
## scale(health) 1.645e-03  3.010e-03  1.230e+05  0.547   0.585
## sex          -1.010e-01  5.283e-03  1.230e+05 -19.126 <2e-16 ***
## age_z         2.162e-01  2.895e-03  1.230e+05  74.696 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) scl(h) sex
## scale(hlth) -0.005
## sex          -0.060  0.056
## age_z         0.001  0.402 -0.022
## optimizer (nlptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

```
emotional_stability_lmer_health <- lmer(
  emotional_stability_scale ~ scale(health) + sex + age_z + (1 | country),
  data      = ts,
  weights = weight
)
summary(emotional_stability_lmer_health)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: emotional_stability_scale ~ scale(health) + sex + age_z + (1 |
##   country)
##   Data: ts
## Weights: weight
##
## REML criterion at convergence: Inf
##
## Scaled residuals:
##   Min      1Q  Median      3Q      Max
```

```
## -9.9152 -0.6009 0.0989 0.6792 6.9902
##
## Random effects:
## Groups Name Variance Std.Dev.
## country (Intercept) 0.03492 0.1869
## Residual 0.92363 0.9611
## Number of obs: 177171, groups: country, 49
##
## Fixed effects:
## Estimate Std. Error df t value Pr(>|t|)
## (Intercept) 3.376e-02 2.693e-02 3.904e+01 1.253 0.218
## scale(health) 1.926e-01 2.572e-03 1.770e+05 74.866 < 2e-16 ***
## sex -2.899e-02 4.583e-03 1.771e+05 -6.327 2.51e-10 ***
## age_z 3.433e-03 2.484e-03 1.772e+05 1.382 0.167
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
## (Intr) scl(h) sex
## scale(hlth) -0.005
## sex -0.088 0.060
## age_z 0.006 0.374 -0.022
## optimizer (nloptwrap) convergence code: 0 (OK)
## Gradient contains NAs
```

With these personality scales (which are not very representative), I observe a
↪ significant and positive relationship of health and openness, as well as of health
↪ and emotional stability. We will continue the following analyses with these two
↪ personality scales.

Mediation analyses - Structural Equation Models

Mediation model for Openness

Path a: Income → Health (the mediator)

Path b: Health → Openness

Path c': Direct effect of Income → Openness controlling for Health

Indirect effect (a × b) and total effect (c)

```
simple_mediation_openness <- "
  health ~ a * income_scale
  openness_scale ~ cprime * income_scale + b * health
  ab := a * b
  total := cprime + ab
"
```

Mediation model for Emotional Stability

```
simple_mediation_emotional_stability <- "
```

Path a: Income → Health (the mediator)

*health ~ a * income_scale*

Path b: Health → Emotional Stability

Path c': Direct effect of Income → Emotional Stability controlling for Health

*emotional_stability_scale ~ cprime * income_scale + b * health*

Indirect effect (a × b) and total effect (c)

```

    ab := a * b
    total := cprime + ab
  "

# Keep only rows where the involved variables are all non-missing
vars_mediation <- c("income_scale", "health", "openness_scale",
  ↪ "emotional_stability_scale")
ts_mediation <- ts[complete.cases(ts[, vars_mediation]), ]

# Fit the models using lavaan
library(lavaan)

fit_mediation_simple_openness <- sem(
  simple_mediation_openness,
  data      = ts_mediation,
  missing   = "fiml"
)
summary(fit_mediation_simple_openness, fit.measures = TRUE, standardized = TRUE)

```

```

## lavaan 0.6.17 ended normally after 1 iteration
##
##      Estimator                      ML
##      Optimization method          NLMINB
##      Number of model parameters          7
##
##      Number of observations          95835
##      Number of missing patterns          1
##
## Model Test User Model:
##
##      Test statistic              0.000
##      Degrees of freedom              0
##
## Model Test Baseline Model:
##
##      Test statistic              7932.265
##      Degrees of freedom              3
##      P-value              0.000
##
## User Model versus Baseline Model:
##
##      Comparative Fit Index (CFI)              1.000
##      Tucker-Lewis Index (TLI)              1.000
##
##      Robust Comparative Fit Index (CFI)              1.000
##      Robust Tucker-Lewis Index (TLI)              1.000
##
## Loglikelihood and Information Criteria:
##
##      Loglikelihood user model (H0)          -250407.779
##      Loglikelihood unrestricted model (H1)    -250407.779
##
##      Akaike (AIC)              500829.559

```

```

## Bayesian (BIC) 500895.851
## Sample-size adjusted Bayesian (SABIC) 500873.605
##
## Root Mean Square Error of Approximation:
##
## RMSEA 0.000
## 90 Percent confidence interval - lower 0.000
## 90 Percent confidence interval - upper 0.000
## P-value H_0: RMSEA <= 0.050 NA
## P-value H_0: RMSEA >= 0.080 NA
##
## Robust RMSEA 0.000
## 90 Percent confidence interval - lower 0.000
## 90 Percent confidence interval - upper 0.000
## P-value H_0: Robust RMSEA <= 0.050 NA
## P-value H_0: Robust RMSEA >= 0.080 NA
##
## Standardized Root Mean Square Residual:
##
## SRMR 0.000
##
## Parameter Estimates:
##
## Standard errors Standard
## Information Observed
## Observed information based on Hessian
##
## Regressions:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## health ~
## incm_sc (a) 0.103 0.001 83.722 0.000 0.103 0.261
## openness_scale ~
## incm_sc (cprm) 0.026 0.001 22.000 0.000 0.026 0.073
## health (b) 0.059 0.003 19.623 0.000 0.059 0.065
##
## Intercepts:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .health 3.194 0.007 463.257 0.000 3.194 3.333
## .openness_scale -0.369 0.012 -31.880 0.000 -0.369 -0.425
##
## Variances:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .health 0.856 0.004 218.901 0.000 0.856 0.932
## .openness_scale 0.745 0.003 218.901 0.000 0.745 0.988
##
## Defined Parameters:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## ab 0.006 0.000 19.105 0.000 0.006 0.017
## total 0.032 0.001 28.041 0.000 0.032 0.090

```

```

fit_mediation_simple_emotional_stability <- sem(
  simple_mediation_emotional_stability,
  data = ts_mediation,
  missing = "fiml"
)

```

```
)
summary(fit_mediation_simple_emotional_stability, fit.measures = TRUE, standardized =
  TRUE)
```

```
## lavaan 0.6.17 ended normally after 1 iteration
##
##      Estimator              ML
##      Optimization method    NLMINB
##      Number of model parameters      7
##
##      Number of observations      95835
##      Number of missing patterns      1
##
## Model Test User Model:
##
##      Test statistic            0.000
##      Degrees of freedom          0
##
## Model Test Baseline Model:
##
##      Test statistic      10825.086
##      Degrees of freedom      3
##      P-value            0.000
##
## User Model versus Baseline Model:
##
##      Comparative Fit Index (CFI)      1.000
##      Tucker-Lewis Index (TLI)        1.000
##
##      Robust Comparative Fit Index (CFI)      1.000
##      Robust Tucker-Lewis Index (TLI)        1.000
##
## Loglikelihood and Information Criteria:
##
##      Loglikelihood user model (H0)      -263028.191
##      Loglikelihood unrestricted model (H1) -263028.191
##
##      Akaike (AIC)            526070.381
##      Bayesian (BIC)           526136.674
##      Sample-size adjusted Bayesian (SABIC) 526114.428
##
## Root Mean Square Error of Approximation:
##
##      RMSEA                    0.000
##      90 Percent confidence interval - lower      0.000
##      90 Percent confidence interval - upper      0.000
##      P-value H_0: RMSEA <= 0.050              NA
##      P-value H_0: RMSEA >= 0.080              NA
##
##      Robust RMSEA              0.000
##      90 Percent confidence interval - lower      0.000
##      90 Percent confidence interval - upper      0.000
##      P-value H_0: Robust RMSEA <= 0.050          NA
```



```
## P-value H_0: Robust RMSEA >= 0.080 NA
##
## Standardized Root Mean Square Residual:
##
## SRMR 0.000
##
## Parameter Estimates:
##
## Standard errors Standard
## Information Observed
## Observed information based on Hessian
##
## Regressions:
## Estimate Std.Err z-value P(>|z|) Std.lv
## health ~
## incm_sc (a) 0.103 0.001 83.722 0.000 0.103
## emotional_stability_scale ~
## incm_sc (cprm) 0.027 0.001 19.630 0.000 0.027
## health (b) 0.186 0.003 54.084 0.000 0.186
## Std.all
##
## 0.261
##
## 0.064
## 0.177
##
## Intercepts:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .health 3.194 0.007 463.257 0.000 3.194 3.333
## .emtnl_stblty_s -0.873 0.013 -66.070 0.000 -0.873 -0.868
##
## Variances:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## .health 0.856 0.004 218.901 0.000 0.856 0.932
## .emtnl_stblty_s 0.970 0.004 218.901 0.000 0.970 0.959
##
## Defined Parameters:
## Estimate Std.Err z-value P(>|z|) Std.lv Std.all
## ab 0.019 0.000 45.429 0.000 0.019 0.046
## total 0.046 0.001 34.440 0.000 0.046 0.111
```

7 Presenting the main results

```
### 1) DESCRIPTIVE TABLE ###

vars_demographics <- c("sex", "age", "social_class", "income_scale",
  "partner", "unemployment", "health")
vars_personality <- c("extraversion_scale", "openness_scale",
  "agreeableness_scale", "conscientiousness_scale",
  "emotional_stability_scale")
```

```

vars_for_summary <- c(vars_demographics, vars_personality)

summary_table_big5 <- data.frame(
  Variable = vars_for_summary,
  Non_NA   = sapply(vars_for_summary, function(v) sum(!is.na(ts[[v]]))),
  Missing  = sapply(vars_for_summary, function(v) sum(is.na(ts[[v]]))),
  Mean     = sapply(vars_for_summary, function(v) round(mean(ts[[v]]), na.rm=TRUE), 2)),
  SD       = sapply(vars_for_summary, function(v) round(sd(ts[[v]]), na.rm=TRUE), 2)),
  Min      = sapply(vars_for_summary, function(v) round(min(ts[[v]]), na.rm=TRUE), 2)),
  Max      = sapply(vars_for_summary, function(v) round(max(ts[[v]]), na.rm=TRUE), 2))
)

rownames(summary_table_big5) <- NULL

desc_table <- kable(
  summary_table_big5,
  format = "latex",
  booktabs = TRUE,
  caption = "Descriptive Statistics",
  label = "tab:desc_bigfive"
) %>%
  kable_styling(latex_options = c("hold_position"), font_size = 8) %>%
  group_rows("Demographics", 1, 7) %>%
  group_rows("Personality Scales", 8, 12)

desc_table

```

Table 2: Descriptive Statistics

Variable	Non_NA	Missing	Mean	SD	Min	Max
Demographics						
sex	224311	123	0.55	0.50	0.00	1.00
age	223506	928	46.21	17.68	15.00	108.00
social_class	843	223591	2.96	0.95	1.00	5.00
income_scale	187462	36972	4.96	2.53	1.00	11.00
partner	224434	0	0.59	0.49	0.00	1.00
unemployment	224434	0	0.08	0.27	0.00	1.00
health	182207	42227	3.73	0.95	1.00	5.00
Personality Scales						
extraversion_scale	29223	195211	0.00	0.72	-1.55	1.36
openness_scale	159801	64633	-0.01	0.88	-0.27	3.84
agreeableness_scale	9458	214976	0.00	0.91	-1.67	1.14
conscientiousness_scale	162663	61771	0.00	1.00	-1.90	1.44
emotional_stability_scale	218931	5503	0.00	1.00	-2.61	1.40

```

### 2) TABLE OF LMER RESULTS (Income, Sex, Age) ####

extract_lmer_coef <- function(model, scale_name) {
  tidy(model, effects = "fixed", conf.int = TRUE) %>%
    filter(term %in% c("(Intercept)", "income_scale_z", "sex", "age_z")) %>%
    mutate(Personality = scale_name) %>%
    select(Personality, term, estimate, conf.low, conf.high, p.value)
}

```

```

res_extraversion <- extract_lmer_coef(extraversion_lmer, "Extraversion")
res_openness <- extract_lmer_coef(openness_lmer, "Openness")
res_agreeableness <- extract_lmer_coef(agreeableness_lmer, "Agreeableness")
res_conscientious <- extract_lmer_coef(conscientiousness_lmer, "Conscientiousness")
res_emotional <- extract_lmer_coef(emotional_stability_lmer, "Emotional Stability")

lmer_results <- dplyr::bind_rows(
  res_extraversion,
  res_openness,
  res_agreeableness,
  res_conscientious,
  res_emotional
)

lmer_table <- kable(
  lmer_results,
  format = "latex",
  digits = 3,
  booktabs = TRUE,
  caption = "Mixed-Effects Model Results (Income, Sex, Age) for Each Personality Scale",
  label = "tab:lmer_results"
) %>%
  kable_styling(latex_options = "hold_position", font_size = 8) %>%
  group_rows("Extraversion", 1, 4) %>%
  group_rows("Openness", 5, 8) %>%
  group_rows("Agreeableness", 9, 12) %>%
  group_rows("Conscientiousness", 13, 16) %>%
  group_rows("Emotional Stability", 17, 20)

lmer_table

```

3) FACETTED PLOT: INTERACTION BETWEEN INCOME AND AGE ON PERSONALITY TRAITS

```

# Define age deciles
age_z_d1 <- quantile(ts$age_z, 0.1, na.rm = TRUE)
age_z_d9 <- quantile(ts$age_z, 0.9, na.rm = TRUE)

# Function to generate predictions
get_trait_income_age_preds <- function(model, trait_name) {
  income_range <- seq(min(ts$income_scale_z, na.rm = TRUE), max(ts$income_scale_z, na.rm
↵ = TRUE), length.out = 100)

  # Mean of sex for controlling
  sex_mean <- mean(ts$sex, na.rm = TRUE)

  # Generate prediction data
  pred_data <- expand_grid(
    income_scale_z = income_range,
    age_z = c(age_z_d1, age_z_d9),
    sex = mean(ts$sex, na.rm = TRUE),
    country = NA
  )
}

```

Table 3: Mixed-Effects Model Results (Income, Sex, Age) for Each Personality Scale

Personality	term	estimate	conf.low	conf.high	p.value
Extraversion					
Extraversion	(Intercept)	0.052	-0.023	0.127	0.170
Extraversion	income_scale_z	0.063	0.055	0.072	0.000
Extraversion	sex	-0.168	-0.183	-0.152	0.000
Extraversion	age_z	-0.248	-0.257	-0.239	0.000
Openness					
Openness	(Intercept)	-0.041	-0.088	0.006	0.088
Openness	income_scale_z	0.066	0.061	0.071	0.000
Openness	sex	0.018	0.009	0.028	0.000
Openness	age_z	-0.023	-0.028	-0.019	0.000
Agreeableness					
Agreeableness	(Intercept)	-0.070	-0.177	0.037	0.192
Agreeableness	income_scale_z	-0.062	-0.083	-0.041	0.000
Agreeableness	sex	0.198	0.162	0.235	0.000
Agreeableness	age_z	0.158	0.136	0.179	0.000
Conscientiousness					
Conscientiousness	(Intercept)	0.091	-0.001	0.182	0.052
Conscientiousness	income_scale_z	-0.068	-0.074	-0.063	0.000
Conscientiousness	sex	-0.110	-0.120	-0.100	0.000
Conscientiousness	age_z	0.217	0.212	0.222	0.000
Emotional Stability					
Emotional Stability	(Intercept)	0.023	-0.037	0.084	0.440
Emotional Stability	income_scale_z	0.100	0.095	0.104	0.000
Emotional Stability	sex	-0.048	-0.057	-0.039	0.000
Emotional Stability	age_z	-0.052	-0.057	-0.047	0.000

```

pred_data$predicted <- predict(model, newdata = pred_data, re.form = NA)

pred_data$AgeGroup <- factor(pred_data$age_z,
                             levels = c(age_z_d1, age_z_d9),
                             labels = c("Young (10th percentile)", "Older (90th
                               ↪ percentile)"))

pred_data$Trait <- trait_name

pred_data
}

# Generate predictions for each trait
pred_extraversion <- get_trait_income_age_preds(extraversion_lmer_interaction,
  ↪ "Extraversion")
pred_openness <- get_trait_income_age_preds(openness_lmer_interaction, "Openness")
pred_agreeableness <- get_trait_income_age_preds(agreeableness_lmer_interaction,
  ↪ "Agreeableness")
pred_conscientiousness <- get_trait_income_age_preds(conscientiousness_lmer_interaction,
  ↪ "Conscientiousness")
pred_emotional_stability <-
  ↪ get_trait_income_age_preds(emotional_stability_lmer_interaction, "Emotional
  ↪ Stability")

# Combine predictions
all_preds <- bind_rows(pred_extraversion, pred_openness, pred_agreeableness,
  ↪ pred_conscientiousness, pred_emotional_stability)

```

```

# Graph ggplot2 of the interactions
graph_interaction_income_age <- ggplot(all_preds, aes(x = income_scale_z, y = predicted,
  ↪ color = AgeGroup, fill = AgeGroup)) +
  geom_line(size = 1) +
  facet_wrap(~ Trait, scales = "fixed") +
  labs(title = "Interaction between Income and Age on Personality Traits",
    x = "Income (standardized)",
    y = "Predicted Personality Score",
    color = "Age Group",
    fill = "Age Group") +
  theme_minimal() +
  theme(plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA),
    plot.title = element_text(hjust = 0.5))

# Display the graph
print(graph_interaction_income_age)

```



4) TABLE OF SEM RESULTS (MEDIATION ANALYSES)

```

sem_openness <- parameterEstimates(fit_mediation_simple_openness, standardized=TRUE)
sem_emotional <- parameterEstimates(fit_mediation_simple_emotional_stability,
  ↪ standardized=TRUE)

```

```

sem_openness$Model <- "Openness"
sem_emotional$Model <- "Emotional Stability"

keep_these <- c(
  "health ~ income_scale",
  "openness_scale ~ income_scale",
  "openness_scale ~ health",
  "emotional_stability_scale ~ income_scale",
  "emotional_stability_scale ~ health",
  "ab",
  "total"
)

sem_combined <- bind_rows(sem_openness, sem_emotional) %>%
  dplyr::filter(
    paste(lhs, rhs, sep=" ~ ") %in% keep_these | lhs %in% c("ab", "total")
  )

sem_readable <- sem_combined %>%
  mutate(
    Dependent = case_when(
      lhs == "ab" ~ "Indirect effect (a*b)",
      lhs == "total" ~ "Total effect (c+ab)",
      lhs == "health" ~ "Mediator: Health",
      lhs == "openness_scale" ~ "Openness",
      lhs == "emotional_stability_scale" ~ "Emotional Stability",
      TRUE ~ lhs
    ),
    Predictor = case_when(
      rhs == "income_scale" ~ "Income",
      rhs == "health" ~ "Health",
      TRUE ~ rhs
    )
  ) %>%
  transmute(
    Model,
    `Path/Effect` = Dependent,
    Predictor,
    Estimate = round(est, 3),
    `95% CI (Low)` = round(ci.lower, 3),
    `95% CI (High)` = round(ci.upper, 3),
    `p-value` = pvalue,
    `Std. Est.` = round(std.all, 3)
  )

kable(
  sem_readable,
  format = "latex",
  digits = 3,
  booktabs = TRUE,
  caption = "SEM Mediation Estimates (Income → Health → Personality). Indirect effect =  

  ↪ a*b, total effect = c + (a*b).",
  label = "tab:sem_results"
) %>%

```

```
kable_styling(latex_options = "hold_position", font_size = 8)
```

Table 4: SEM Mediation Estimates (Income $\rightarrow$ Health $\rightarrow$ Personality). Indirect effect = $a*b$, total effect = $c + (a*b)$.

Model	Path/Effect	Predictor	Estimate	95% CI (Low)	95% CI (High)	p-value	Std. Est.
Openness	Mediator: Health	Income	0.103	0.100	0.105	0	0.261
Openness	Openness	Income	0.026	0.024	0.028	0	0.073
Openness	Openness	Health	0.059	0.053	0.065	0	0.065
Openness	Indirect effect ($a*b$)	$a*b$	0.006	0.005	0.007	0	0.017
Openness	Total effect ($c+ab$)	$cprime+ab$	0.032	0.030	0.034	0	0.090
Emotional Stability	Mediator: Health	Income	0.103	0.100	0.105	0	0.261
Emotional Stability	Emotional Stability	Income	0.027	0.024	0.029	0	0.064
Emotional Stability	Emotional Stability	Health	0.186	0.179	0.193	0	0.177
Emotional Stability	Indirect effect ($a*b$)	$a*b$	0.019	0.018	0.020	0	0.046
Emotional Stability	Total effect ($c+ab$)	$cprime+ab$	0.046	0.043	0.048	0	0.111

```
### 5) GRAPH ILLUSTRATING THE SEM ###
```

```
# For Openness
```

```
if (exists("fit_mediation_simple_openness")) {
```

```
  # Income  $\rightarrow$  Health : 0.26
```

```
  # Health  $\rightarrow$  Openness : 0.065
```

```
  # Income  $\rightarrow$  Openness (effet direct) : 0.073
```

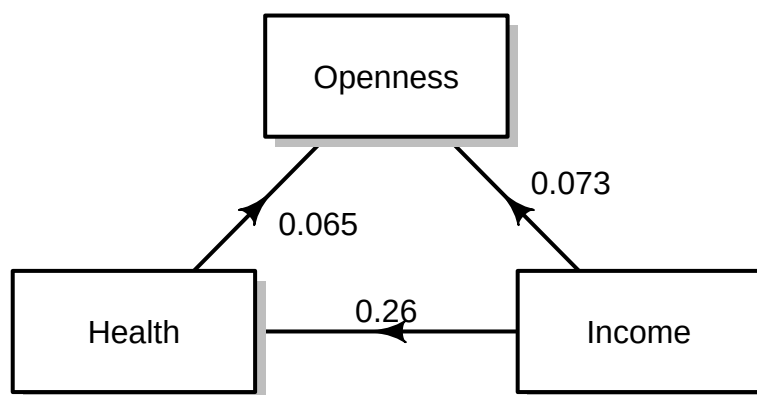
```
  data_opp <- c(0, "0.065", "0.073",
               0, 0, "0.26",
               0, 0, 0)
```

```
  M_opp <- matrix(data_opp, nrow = 3, ncol = 3, byrow = TRUE)
```

```
  plotmat(M_opp, pos = c(1, 2),
           name = c("Openness", "Health", "Income"),
           box.type = "rect",
           box.size = 0.12,
           box.prop = 0.5,
           curve = 0,
           main = "Mediation model for Openness")
```

```
}
```

Mediation model for Openness



```

# For Emotional Stability
if (exists("fit_mediation_simple_emotional_stability")) {
  library(diagram)

  # Income → Health : 0.26
  # Health → Emotional Stability : 0.177
  # Income → Emotional Stability (effet direct) : 0.064
  data_es <- c(0,      "0.177", "0.064",
               0,      0,      "0.26",
               0,      0,      0)

  M_es <- matrix(data_es, nrow = 3, ncol = 3, byrow = TRUE)

  plotmat(M_es, pos = c(1, 2),
           name = c("Emotional Stability", "Health", "Income"),
           box.type = "rect",
           box.size = 0.12,
           box.prop = 0.5,
           curve = 0,
           main = "Mediation model for Emotional Stability")
}

```


Mediation model for Emotional Stability

